

Jesus Andres Ruiz

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Education

University of Texas at San Antonio <i>Bachelor of Science in Mechanical Engineering GPA: 3.6</i>	San Antonio, TX May 2028
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Skills

- Software & Tools:** SolidWorks, Fusion 360 (CAD + Simulation), FEA, MATLAB, Microsoft Office
- Programming / Embedded:** Arduino, Python, Git/GitHub, ROS (learning)
- Manufacturing & Design:** DFM, GD&T, Design Iteration, Rapid Prototyping, Failure/Root Cause Analysis
- Fabrication & Tools:** FDM & Continuous Carbon Fiber Printing, Wiring/Soldering, Mig/Tig Welding, CNC Machining
- Materials & Testing:** 6061-T6 Aluminum, Carbon Fiber Composites, Data Acquisition, Instrumentation, Schlieren Imaging

Relevant Experience

Intelligent Systems and Bionics Engineering Lab <i>Undergraduate Research Assistant</i>	San Antonio, TX May 2026 – Present
- Led fabrication of the OpenExo hip module, an open-source wearable exoskeleton, working closely with graduate students to fabricate and assemble hardware across 3D printing and machining. - Diagnosed and resolved 3D printer hardware/adhesion issues across 2 printer platforms (FDM and continuous carbon fiber), restoring consistent prototyping output.	
UTSA Hypersonics Lab <i>Undergraduate Research Volunteer</i>	San Antonio, TX January 2026 – May 2026
- Fabricated precision diaphragms in the machine shop for use in a Mach 7.2 Ludwig Tube wind tunnel, supporting high-speed flow experiments and ensuring reliable burst pressures for test runs. - Assisted researchers during hypersonic wind tunnel testing, helping prepare instrumentation, monitor runs, and collect experimental data from the wind tunnel. - Set up and aligned laser-based Schlieren imaging systems to visualize shock waves and boundary-layer structures in hypersonic flow experiments.	
UTSA Formula Society of Automotive Engineers <i>Mechanical Engineer - Chassis / Brakes</i>	San Antonio, TX August 2025 – Present
- Machined a set of aluminum spacers on the mill and lathe to set caliper-to-rotor alignment on the 2025 competition vehicle, holding thickness tolerances required for even pad contact and consistent clearance. - Modeled brake-system components in SolidWorks and produced dimensioned engineering drawings with GD&T callouts, applying design-for-manufacturing principles to keep parts producible on in-house shop equipment. - Collaborated with a 5-person chassis and brakes sub-team to iterate designs under FSAE competition constraints, supporting fitment checks and assembly during vehicle build.	

Project Experience

Open Exo Hip Module <i>Undergraduate Research Assistant – Project Lead</i>	San Antonio, TX May 2026 – Present
- Re-designed hip module components in Fusion 360 and validated redesigns using Fusion 360 static stress simulation following structural failure analysis of cracked cuffs, sliders, and struts on early prototypes. - Manufactured parts via 3D printing (FDM, continuous carbon fiber), CNC machining, and outsourced cutting; corrected a 200-micron shoulder screw bore tolerance issue caused by anodizing buildup before final assembly. - Iterated the hip abductor through 3 design/print revisions before finalizing it as a machined, anodized 6061-T6 aluminum part.	
Bowden Cable Transmission – Ankle Actuation <i>Undergraduate Research Assistant</i>	San Antonio, TX May 2026 – Present
- Prototyped a Bowden cable transmission enabling remote ankle actuation, reducing distal leg mass and metabolic cost of walking. - Designed, printed, and tested 2 TPU cable insert geometries (tapered vs. straight) to optimize routing and fit for cables.	